Portuguese Customer Review Score Classification

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***Abstract*—Customer reviews provide information about purchasing experiences, but the numerical rating associated with a written comment is not always explicit. This project develops a multiclass machine-learning pipeline to predict one-to-five-star ratings from Portuguese reviews in the Brazilian E-Commerce Public Dataset by Olist. After data preparation, 40,634 reviews are divided into training, validation, and test subsets. Four approaches are examined in sequence: TF-IDF with multinomial logistic regression, a CNN with frozen fastText embeddings, a CNN with learned embeddings, and an ordinal version of the learned-embedding CNN. The learned Embedding CNN is selected because it combines the highest validation macro-F1 among the four argmax models with a low mean absolute error and greater recognition of intermediate ratings. Its validation accuracy is 68.38%, macro-F1 is 0.4511, and MAE is 0.4803 stars. On 8,127 held-out test reviews, the model obtains approximately 68% accuracy, 0.45 macro-F1, and an MAE of 0.4924 stars. These results support approximate rating estimation while showing substantial limitations in distinguishing two-, three-, and four-star reviews.**

Keywords—natural language processing, Portuguese reviews, TF-IDF, logistic regression, fastText, convolutional neural network, word embeddings, ordinal classification, mean absolute error.

I. Introduction

Online customer reviews contain information about product quality, delivery time, missing items, and the overall purchasing experience. Together with a numerical score, these comments allow other customers and sellers to understand how a transaction was perceived. However, interpreting a large collection of written reviews manually requires time and can produce inconsistent judgments.

Machine learning provides a way to analyze review text and estimate the rating associated with it. In this project, the review title and comment are used as the only predictive inputs. The task contains five ordered categories, from one star to five stars. This order is relevant because predicting four stars for a five-star review represents a smaller rating error than predicting one star.

The analysis begins with a TF-IDF baseline and then examines convolutional models with different word representations and training objectives. The comparison considers both exact classification and the distance between predicted and observed ratings. This provides a more complete interpretation than accuracy alone, particularly when the target classes are imbalanced.

II. Objective

The objective of this project is to develop and evaluate a multiclass model that predicts the review score from Portuguese customer comments. The specific goals are to prepare the text data, compare four modeling approaches under a common partition, examine performance for every score, and select a model that provides useful approximate predictions while maintaining recognition across the five categories. The final model is evaluated on a held-out test set and can be reused through a text-input interface without retraining.

III. Background and Theoretical Foundations

A. Text Representation and Word Embeddings

A machine-learning model requires numerical inputs. TF-IDF represents a review through weighted word frequencies, whereas integer vectorization assigns a vocabulary index to each token. An integer index identifies a word; its numerical magnitude does not represent sentiment or semantic similarity. An embedding layer converts each index into a dense vector whose coordinates can be learned from the rating-prediction task.

Pretrained fastText vectors offer another initialization strategy. They provide word representations learned from a larger language corpus. In the experiment, Portuguese vectors are loaded from a compressed text file and kept frozen. This differs from learning the embedding directly from the Olist reviews. Although fastText uses subword information when creating its representations, this implementation loads a fixed word-vector table and does not generate vectors dynamically for unseen words.

B. Convolutional Neural Networks for Text

A one-dimensional convolution applies the same learned filter to successive groups of word vectors. With a kernel width of three, each filter examines a local three-token window. The filter can learn patterns associated with positive or negative experiences, although the meaning of individual filters is not directly established by the experiment. Global max pooling retains the strongest activation of each filter anywhere in the review, and a final classifier converts these features into rating probabilities.

C. Ordered Review Scores

The ratings represent ordered categories rather than unrelated labels. Nevertheless, ordinary multiclass cross-entropy treats the correct class as the target without explicitly encoding the distance between ratings. An ordinal objective can instead express a rating through cumulative questions, such as whether it is greater than one, two, three, or four stars. Both approaches are evaluated because

classification quality and rating-distance error don't improve together.

IV. Dataset and Data Preparation

A. Dataset Source and Input Variables

The data come from the Brazilian E-Commerce Public Dataset by Olist, available on Kaggle [1]. This study uses the order-review table. Other tables, including customers, sellers, geolocation, payments, products, and order items, are not joined because the intended prediction is based on the written review itself. Including additional transaction attributes would define a different experiment.

Field	Role
review_score	Target, from 1 to 5
review_comment_title	Optional text input
review_comment_message	Required comment body
review_id / order_id	Identifiers for cleaning
Review timestamps	Excluded from predictors

Table 1. Roles of the review-table fields.

The original review table contains 99,224 rows and seven columns. There are 40,977 non-null comment bodies and 11,568 non-null titles. Consequently, many transactions contain a score without a written comment. Such observations cannot support the text-only prediction task in its present form.

V. Exploratory Data Analysis

A. Data Quality Assessment

The preparation notebook checks duplicated review identifiers and repeated order-comment combinations. There are 814 occurrences beyond the first occurrence of each review identifier. Keeping the first occurrence reduces the table to 98,410 rows. The cleaning procedure also removes repeated order-comment combinations, excludes missing or blank comment bodies, and replaces missing titles with empty strings. The final modeling dataset contains 40,634 reviews. A review with only a title is excluded because the procedure requires a nonempty body.

B. Target Class Distribution

Score	Count	Percentage
1	8,640	21.26%
2	2,118	5.21%
3	3,528	8.68%
4	5,935	14.61%
5	20,413	50.24%

Table 2. Score distribution after cleaning.

The distribution shows a concentration at the endpoints. Five-star reviews account for approximately half of the dataset, while two-star reviews form the smallest class. A model that favors the most common ratings can therefore obtain a moderate overall accuracy while failing to recognize intermediate scores. Class-level recall and macro-F1 are necessary to reveal this behavior.

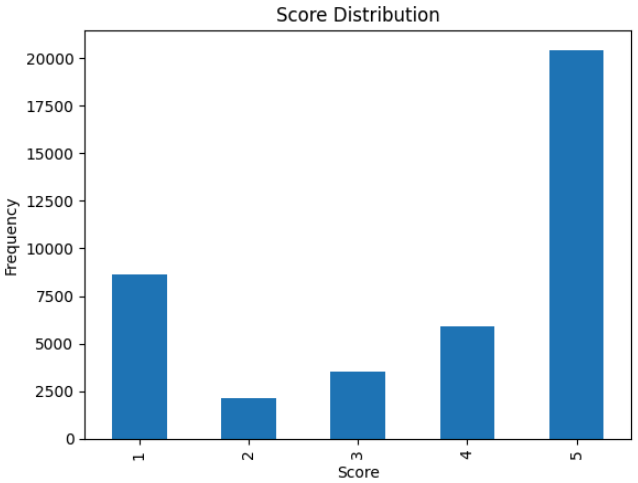


Figure 1. Distribution of the 40,634 cleaned reviews.

C. Text and Label Ambiguity

The previously inspected errors include negative descriptions labeled with two stars that the model predicts as one star. Statements about undelivered products, missing items, or poor quality indicate dissatisfaction, but may not specify whether the author intended one or two stars. Similar ambiguity can occur between four and five stars. These examples suggest overlap in the language used across adjacent ratings; they do not establish that the labels are incorrect or that a performance ceiling has been reached.

VI. Data Preprocessing

A. Text Construction and Label Encoding

The optional title and the comment body are concatenated into a single string. TextVectorization lowercases the text and removes punctuation using its default standardization. Whitespace separates tokens. The vocabulary is adapted using the training texts only and is limited to 15,000 entries. Accented Portuguese words remain part of the vocabulary. For the CNNs, reviews are represented as integer sequences of length 150, with padding or truncation as necessary. Scores are encoded internally as 0–4 and converted back to 1–5 for interpretation.

B. Training, Validation, and Test Partition

Two stratified splits with random_state=42 produce 26,005 training reviews, 6,502 validation reviews, and 8,127 test reviews, corresponding approximately to 64%, 16%, and 20% of the data. Stratification preserves the rating distribution. The training subset is used to adapt the

vocabulary and fit model parameters. The validation subset supports early stopping and model comparison. The test subset is used for the final evaluation of the selected Embedding CNN.

The split is performed at the review-row level. It does not explicitly group reviews by customer or identical text, so independence across these groups is not guaranteed. Future evaluation could use grouped or chronological partitions to investigate generalization to different customers or later transactions.

C. Class Imbalance Treatment

The CNN experiments use the square root of balanced class weights, which increases the contribution of less frequent classes without applying the full inverse-frequency correction. For class k with n_k training observations and N training reviews, the weight is given by

$$w_k = \sqrt{[N / (5n_k)]}$$

The approximate weights for scores one through five are 0.9699, 1.9585, 1.5177, 1.1702, and 0.6310. These weights are computed from the training labels. No synthetic reviews are introduced. The TF-IDF baseline uses its original unweighted configuration, so the comparison reflects complete experimental configurations rather than an isolated comparison of representations.

VII. Classification Methodology

A. TF-IDF and Multinomial Logistic Regression

The first model transforms each review into a TF-IDF vector. Under the Keras TextVectorization convention used here, the inverse document frequency for term t can be expressed as follows, where N is the number of training documents and df(t) is the number containing that term.

$$\text{tf}(t, d) = \frac{\textit{number of times t appears in a document}}{\textit{total number of terms in d}}$$

$$\text{idf}(t) = \log[\, N / (1 + \text{df})]$$

$$\text{tf - idf}(t,d) = \text{tf}(t, d) \cdot \text{idf}(t)$$

A dense layer maps this vector directly to five logits, followed by softmax. With no hidden layer, this architecture implements multinomial logistic regression. It contains 75,005 parameters. Its computational simplicity makes it a useful baseline, although the bag-of-words representation does not preserve token order.

$$p_k = \exp(z_k) / \Sigma_{k=1}^{5} \exp(z_k),$$

$$z = Wx + b$$

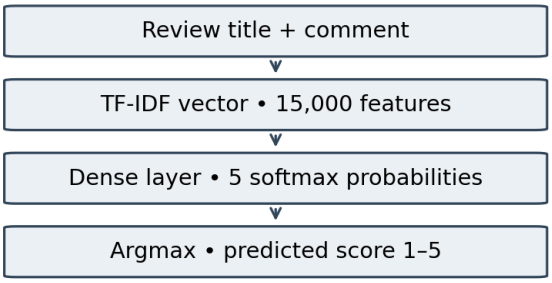


Figure 2. TF-IDF baseline architecture.

B. Fast Text CNN

The second model uses the same integer vocabulary to retrieve pretrained 300-dimensional Portuguese fastText vectors [2]. The compressed vector file is read sequentially, retaining only the vectors needed for the training vocabulary. A total of 11,620 of 14,998 non-special vocabulary entries are matched, corresponding to 77.5% coverage. Unmatched entries receive small random vectors, and the padding vector is initialized to zero. The embedding remains frozen during training.

The embedding output passes through a Conv1D layer with 128 filters, kernel size three, and ReLU activation. Global max pooling reduces the sequence to 128 features, dropout uses a rate of 0.3, and a dense softmax layer returns five probabilities. The model contains 4,615,973 parameters, of which 115,973 are trainable. Freezing the embeddings reduces the number of updated parameters, but also prevents unmatched random vectors from adapting to the task.

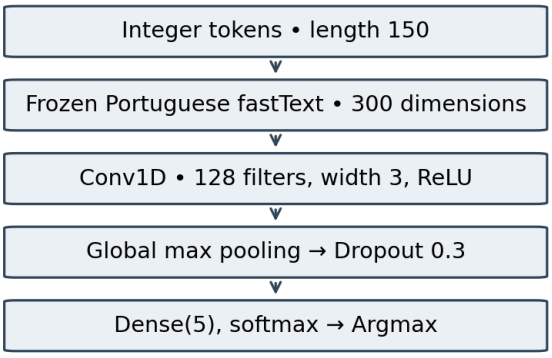


Figure 3. CNN with frozen fastText word vectors.

C. Embedding CNN

The third model learns its embedding table directly from the training reviews. Each token receives a 128-dimensional vector. The convolution, pooling, dropout, and output layers follow the same design as the fastText model. In this setting, the embedding coordinates are updated together with the convolution and classifier, allowing the representation to reflect the rating-prediction objective.

$$e_i = E[\text{token}_i]$$

$$h_{i,k} = \text{ReLU}(\langle F_k, E_{i:i+2} \rangle + b_k)$$

$$g_j = \max_i h_{i,j}$$

Here, E is the embedding matrix, F_j is the j-th convolution filter, and g_j is its pooled activation. The model contains 1,969,925 trainable parameters. Most parameters belong to the 15,000-by-128 embedding table. The convolution contains 49,280 parameters, and the final dense layer contains 645 parameters.

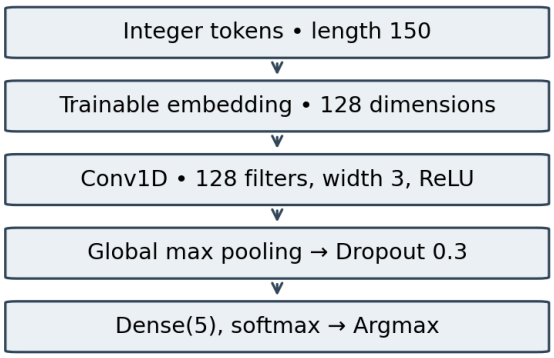


Figure 4. Selected CNN with learned word embeddings.

D. Ordinal Embedding CNN

The fourth model uses a fresh learned-embedding CNN with a five-class softmax output. During training, a fixed, nontrainable transformation converts the five probabilities into four cumulative probabilities. For a rating r on the 1–5 scale, the threshold targets and probabilities are defined by

$$t_k = 1[r > k], \quad q_k = \Sigma_{j=k+1}^{5} p_j, \quad k = 1,\ldots,4$$

For example, a three-star review has threshold targets [1, 1, 0, 0]. Binary cross-entropy compares the cumulative probabilities with these targets. The same class-derived sample weights are applied. Because each cumulative probability is a sum of the softmax probabilities, their ordering is consistent by construction. At inference, the shared five-class predictor returns star probabilities and the most probable star is selected.

$$L_{ordinal} = -(1/4) \Sigma_{k=1}^{4} s_k$$

$$s_k = t_k \ln(q_k) + (1-t_k) \ln(1-q_k)$$

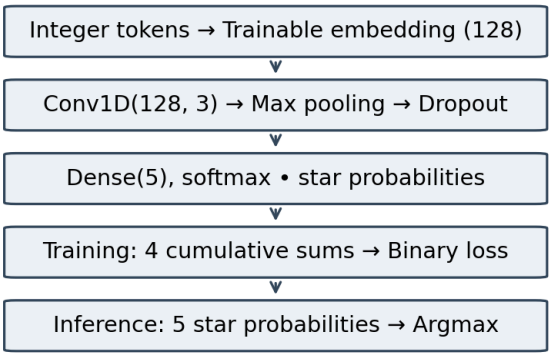


Figure 5. Ordinal training and star-score inference.

E. Optimization and Early Stopping

The models are trained using Adam, batches of 64 reviews, and a maximum of 50 epochs. The ordinary classifiers use sparse categorical cross-entropy. For the class-weighted CNNs, the contribution of each training example is multiplied by its class weight.

$$L = -(1/N) \Sigma_{i=1}^{N} w_{\gamma i} \ln(p_{i,\gamma i})$$

Early stopping monitors validation loss with a patience of three epochs and restores the weights from the best epoch. Thus, 50 epochs is an upper limit rather than a required duration. The selected Embedding CNN reaches its lowest validation loss at epoch two and stops after epoch five. The ordinal model also restores its second-epoch weights.

F. Evaluation Metrics

Accuracy measures the proportion of reviews assigned the exact observed rating. Precision indicates how many predictions of a class are correct, whereas recall indicates how many actual examples of that class are recovered. F1 combines precision and recall through their harmonic mean.

$$Accuracy = (1/N) \Sigma_{i=1}^{N} 1[\hat{y}_i = y_i]$$

$$Precision_k = TP_k / (TP_k + FP_k)$$

$$Recall_k = TP_k / (TP_k + FN_k)$$

$$F1_k = 2TP_k / (2TP_k + FP_k + FN_k)$$

$$Macro\text{-}F1 = (1/5) \Sigma_{k=1}^{5} F1_k$$

Macro-F1 gives equal importance to each score, regardless of its frequency. It is therefore useful for identifying weak intermediate-class performance that can be hidden by the majority classes. MAE measures rating distance in stars, while within-one-star accuracy counts predictions whose absolute error is at most one.

$$MAE = (1/N) \Sigma_{i=1}^{N} |y_i - \hat{y}_i|$$

$$W_1 = (1/N) \Sigma_{i=1}^{N} 1[|y_i - \hat{y}_i| \le 1]$$

A low MAE and a modest macro-F1 can coexist. Predicting one star for a two-star review is an incorrect class prediction but contributes only one star of absolute error. Correct predictions for the frequent endpoint classes further reduce overall MAE. Consequently, MAE alone does not demonstrate reliable recognition of every rating.

G. Baseline Model Results

The TF-IDF baseline achieves a validation accuracy of 69.06%, a macro-F1 of approximately 0.41, and an MAE of 0.4882 stars. Its one- and five-star F1 scores are approximately 0.76 and 0.85, respectively. In contrast, the two-, three-, and four-star F1 scores are approximately 0.10, 0.16, and 0.19. These results establish that useful rating-related information is present in the text, but the baseline favors the endpoint classes.

H. CNN Model Improvement and Comparison

Model	Accuracy	Macro-F1	MAE
TF-IDF	0.6906	≈0.41	0.4882
fastText CNN	0.6912	0.4266	0.4843
Embedding CNN	0.6838	0.4511	0.4803
Ordinal CNN	0.6916	0.4306	0.4780

Table 3. Validation comparison using argmax predictions.

The frozen fastText model slightly improves the baseline macro-F1 and MAE, but its two-star recall remains 0.0531. Pretrained word vectors therefore do not automatically resolve the distinction between closely related rating categories. Partial vocabulary coverage and the frozen representation are possible contributing factors, although the present experiments do not isolate their individual effects.

The learned Embedding CNN achieves the highest macro-F1 among the four configurations under the common argmax rule. Its accuracy is lower than the other models, but it improves the balance between exact category recognition and rating distance. Its two-star recall rises to 0.1298, compared with 0.0531 for fastText. This remains a weak absolute result, but represents greater recognition of the rare class.

Model	2-star recall	Within 1 star
Fast_Text_CNN	0.0531	0.8834
Embedding CNN	0.1298	0.8902
Ordinal CNN	0.0059	0.8867

Table 4. Complementary validation metrics.

The ordinal model obtains the lowest validation MAE, 0.4780 stars, and the highest accuracy, 0.6916, among these configurations. However, its two-star recall falls to 0.0059, meaning that only approximately two of the 339 actual two-star reviews are correctly identified. The ordinal objective improves average rating distance slightly while reducing intermediate-class recognition. It is therefore not selected for the final application.

I. Training Behavior

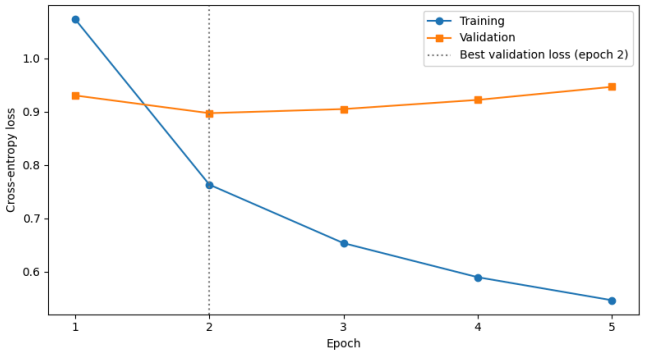


Figure 6. Loss history of the selected Embedding CNN.

The validation loss from the Embedding CNN model decreases from 0.9029 to 0.8745 during the first two epochs and then increases, reaching 1.0340 at epoch five. Meanwhile, training accuracy rises from 0.6475 to 0.8168. This pattern indicates overfitting after the best validation epoch. Figure 6 shows weighted training loss and unweighted validation loss, so their absolute levels are not directly comparable; their trends support the early-stopping decision.

J. Prediction Rule Analysis

The most probable star is the category with the largest individual probability. The median star is the first category whose cumulative probability reaches 0.5. They are alternative decision rules applied to the same model outputs.

$$\hat{y}_{mode} = \arg\max_k p_k$$

$$\hat{y}_{median} = \min\{k : \sum_{i=1}^{k} p_i \geq 0.5\}$$

For the selected CNN, the median rule raises validation macro-F1 from 0.4511 to 0.4584 and two-star recall from 0.1298 to 0.3805. However, accuracy decreases from 0.6838 to 0.6203 and MAE increases from 0.4803 to 0.5043. The argmax rule is retained for the final evaluation. Although a distributional median minimizes absolute loss under that predicted distribution, estimated probabilities can be imperfect, particularly after class weighting, so empirical MAE need not improve.

K. Final Model Selection and Test Evaluation

The Embedding CNN is selected as the preferred model for the present objective because it provides the strongest validation macro-F1 among the four argmax configurations,

the highest within-one-star accuracy among the CNNs, and an MAE close to the best observed result. Its validation MAE is only 0.0023 stars higher than that of the ordinal model, while its two-star recall is substantially greater. This is a selection based on complementary criteria rather than a claim that the model leads every metric.

After model selection, the original Embedding CNN is evaluated on 8,127 test reviews using the same argmax rule. The test MAE is 0.4924326319675157 stars, compared with 0.48031374961550294 on validation. The difference is approximately 0.0121 stars. The test classification report gives accuracy of approximately 0.68, macro-F1 of approximately 0.45, and weighted F1 of approximately 0.65. These classification values were printed to two decimal places and should not be interpreted as more precise estimates.

Score	Precision	Recall	F1	Support
1	.68	.86	.76	1,728
2	.20	.13	.16	423
3	.28	.24	.26	706
4	.35	.19	.25	1,187
5	.81	.88	.84	4,083

Table 5. Final Embedding CNN test classification report.

The test results preserve the pattern observed during validation. The model recognizes one- and five-star reviews relatively well, while recall for scores two, three, and four remains low. An MAE below half a star means that the average absolute rating error is approximately 0.49 stars across all test examples. It does not mean that every prediction is within half a star. The held-out test within-one-star rate is not available from the saved output and is not inferred from validation results.

For perspective, always predicting five stars on this test distribution would yield approximately 50.24% accuracy and an MAE of 1.3264 stars. The selected model improves substantially over that constant prediction. Nevertheless, the low intermediate-class F1 scores limit its suitability for decisions requiring accurate distinctions between neighboring ratings.

L. Model Reuse and User Interface

The trained model is saved in the Keras format with its text vectorization and learned parameters. A simple interface accepts a Portuguese title and comment, combines them into the same input format, and displays the most probable rating with the five output probabilities. Keeping preprocessing inside the saved model helps preserve the vocabulary mapping during inference. The displayed probabilities are model scores and are not established as calibrated confidence estimates.

VIII. Conclusion

This project compares four approaches to Portuguese review-score prediction using a shared data preparation procedure and stratified partition. The TF-IDF baseline captures useful sentiment information but performs poorly on intermediate ratings. Frozen fastText vectors provide a modest change in performance, while learning task-specific embeddings yields a better macro-F1 under the common argmax prediction rule. The ordinal objective slightly reduces MAE but almost eliminates correct recognition of two-star reviews.

The Embedding CNN is selected because it offers the preferred balance across the evaluation criteria. Its test MAE of 0.4924 stars supports its use as an approximate rating estimator, and its test results are broadly consistent with validation. At the same time, a macro-F1 of approximately 0.45 and low intermediate-class recall show that fine-grained rating classification remains difficult. The reported results should therefore be interpreted as useful progress on approximate prediction, with clear class-specific limitations.

The comparison is based on individual saved training runs rather than repeated-seed estimates, so small metric differences do not establish statistical superiority. Other limitations include review-level splitting, exclusion of reviews without comment bodies, truncation to 150 tokens, and possible ambiguity between adjacent scores. Future work could investigate grouped evaluation, contextual Portuguese language models, probability calibration, and repeated runs. Further model development should use validation data, with a fresh held-out evaluation if the current test results influence new design choices.

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